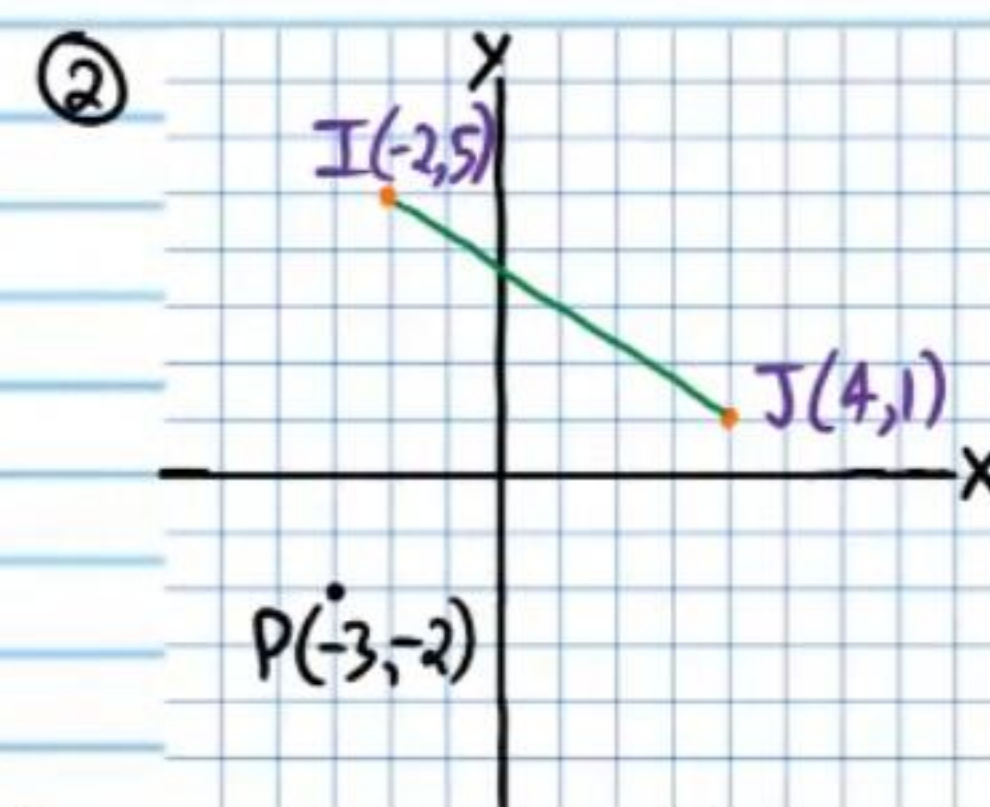
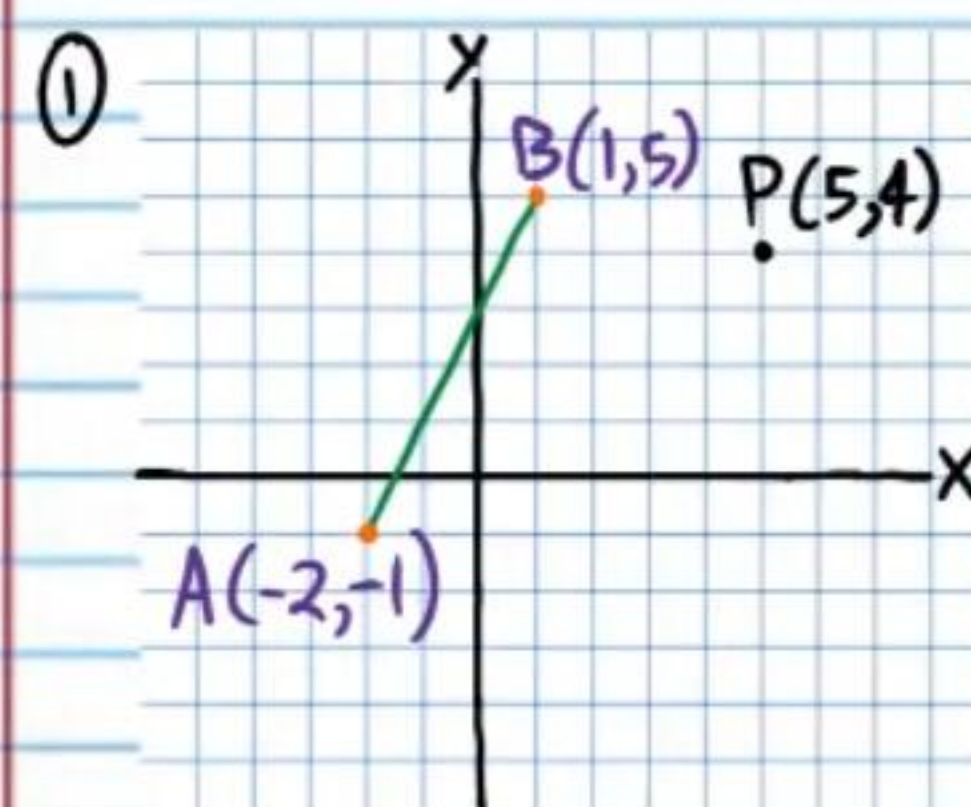


Finding the Coordinates of Points of Concurrency Generated by Triangles

Slopes of Perpendicular Line Segments

If the slopes of two line segments are opposite reciprocals, then the measure of the angle between the line segments is 90° .

For each problem below, find an equation of the line perpendicular to the segment graphed and running through point P.



I) $M = \frac{y_2 - y_1}{x_2 - x_1}$

$M_{AB} = \frac{5 - (-1)}{1 - (-2)} = \frac{6}{3} = 2$

II) $M_{\perp AB} = -\frac{1}{2}$

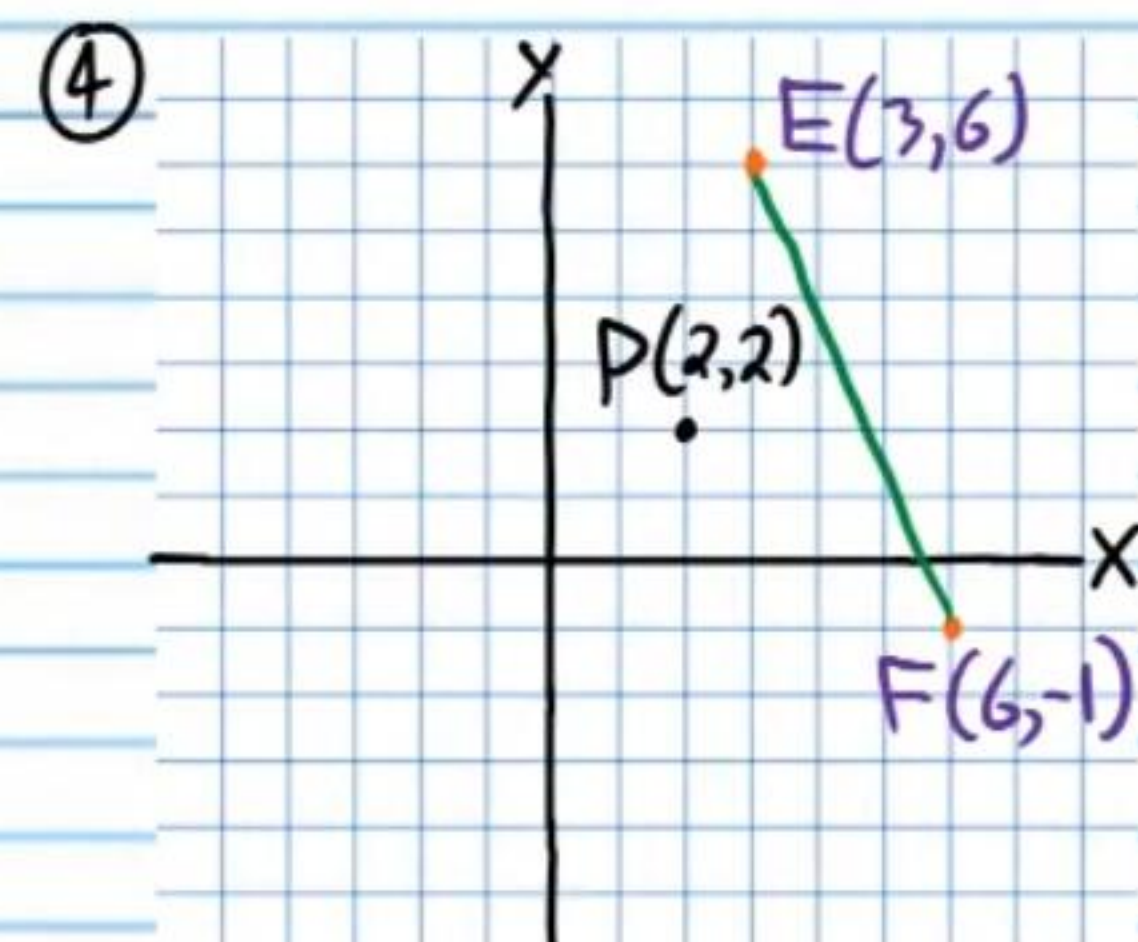
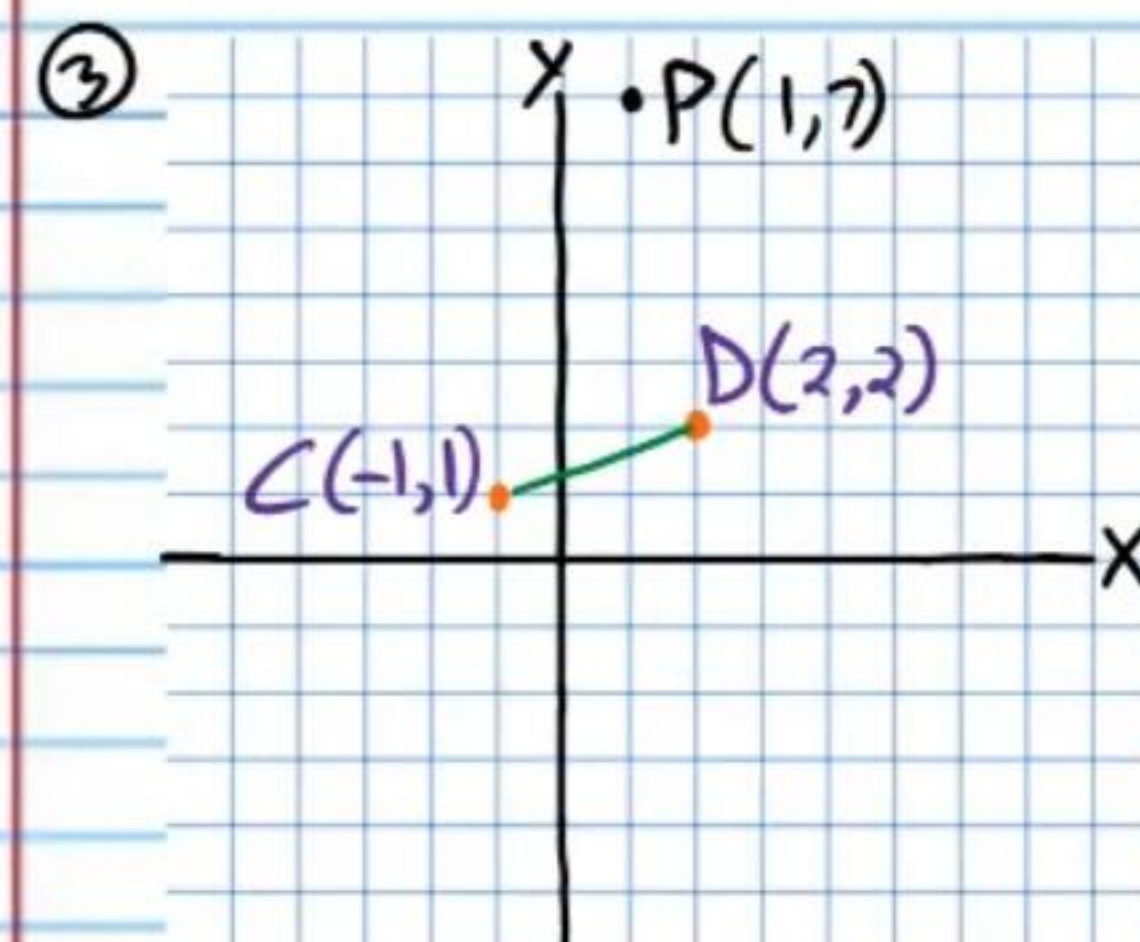
III) $y - y_p = m(x - x_p)$

$y - 4 = -\frac{1}{2}(x - 5)$

I) $M_{IJ} = \frac{1 - 5}{4 - (-2)} = \frac{-4}{6} = -\frac{2}{3}$

II) $M_{\perp IJ} = \frac{3}{2}$

III) $y + 2 = \frac{3}{2}(x + 3)$



I) $M_{CD} = \frac{2 - 1}{2 - (-1)} = \frac{1}{3}$

II) $M_{\perp CD} = -\frac{3}{1} = -3$

III) $y - 7 = -3(x - 1)$

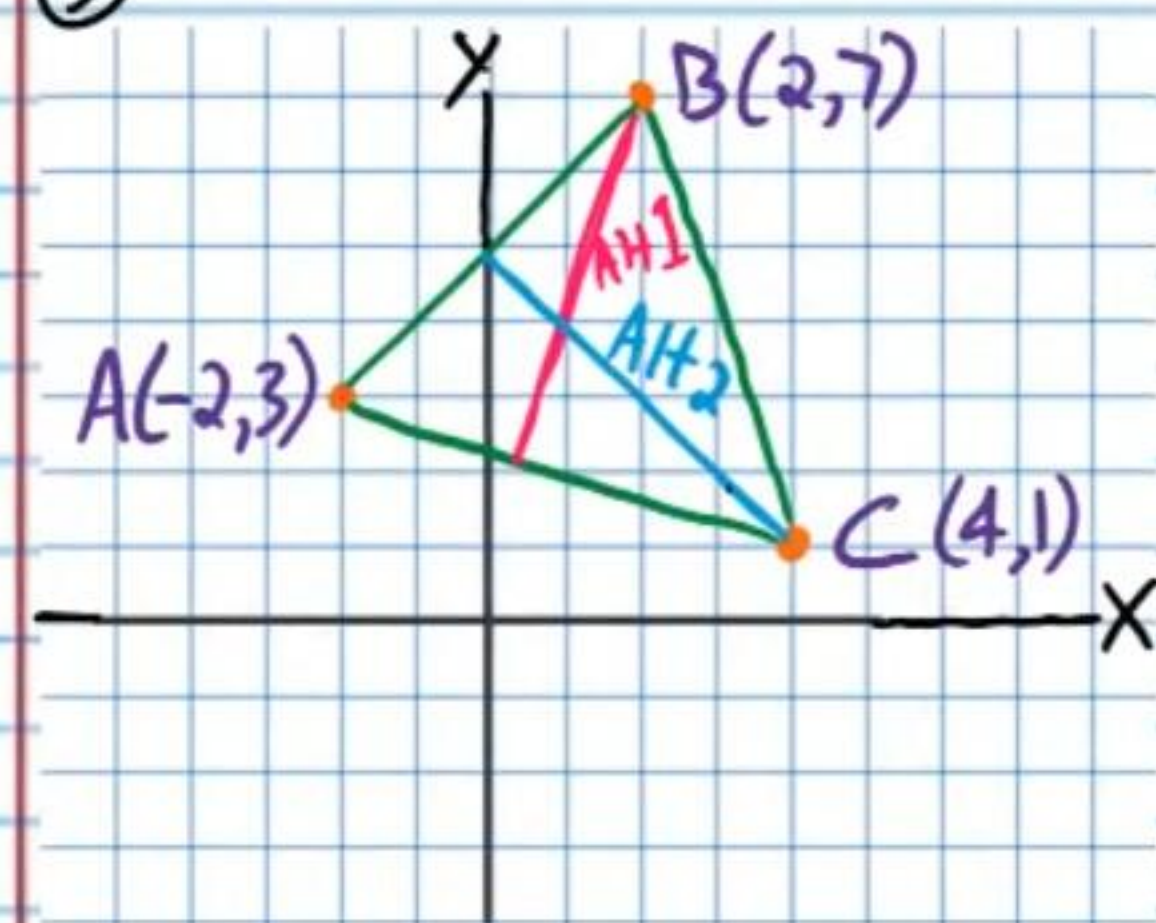
I) $M_{EF} = \frac{-1 - 6}{6 - 3} = -\frac{7}{3}$

II) $M_{\perp EF} = \frac{3}{7}$

III) $y - 2 = \frac{3}{7}(x - 2)$

Given the graphs of the triangles below, find the coordinates of the orthocenters.

⑤



Line Containing Altitude 1

I) $m_{AC} = \frac{1-3}{4-(-2)} = \frac{-2}{6} = -\frac{1}{3}$

II) $m_{\perp AC} = 3$

III) $y-7 = 3(x-2)$

Line Containing Altitude 2

I) $m_{AB} = \frac{7-3}{2-(-2)} = \frac{4}{4} = 1$

II) $m_{\perp AB} = -1$

III) $y-1 = -(x-4)$

Intersection Point

$$\begin{cases} y-7 = 3(x-2) \\ y-1 = -(x-4) \end{cases}$$

$$\begin{cases} y = 3(x-2) + 7 \\ y-1 = -(x-4) \end{cases}$$

$$3(x-2) + 7 - 1 = -(x-4)$$

$$3x - 6 + 7 - 1 = -x + 4$$

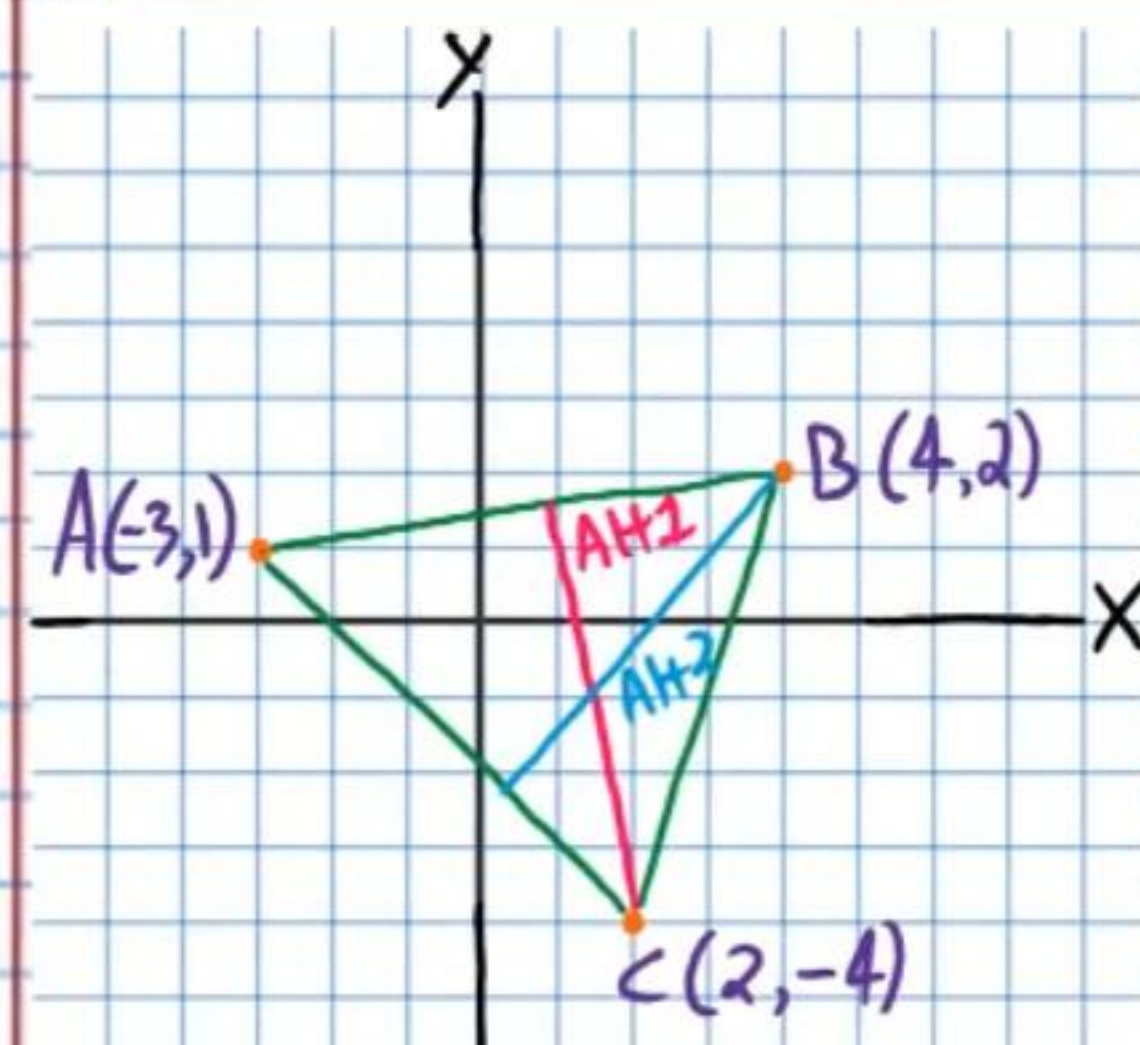
$$x = 1$$

$$y = 3(1-2) + 7$$

$$y = 4$$

Orthocenter: (1, 4)

⑥



Line Containing Altitude 1

I) $m_{AB} = \frac{2-1}{4-(-3)} = \frac{1}{7}$

II) $m_{\perp AB} = -7$

III) $y+4 = -7(x-2)$

Line Containing Altitude 2

I) $m_{AC} = \frac{-4-1}{2-(-3)} = \frac{-5}{5} = -1$

II) $m_{\perp AC} = 1$

III) $y-2 = 1(x-4)$

Intersection Point

$$\begin{cases} y+4 = -7(x-2) \\ y-2 = 1(x-4) \end{cases}$$

$$\begin{cases} y = -7(x-2) - 4 \\ y-2 = 1(x-4) \end{cases}$$

$$-7(x-2) - 4 - 2 = 1(x-4)$$

$$-7x + 14 - 4 - 2 = x - 4$$

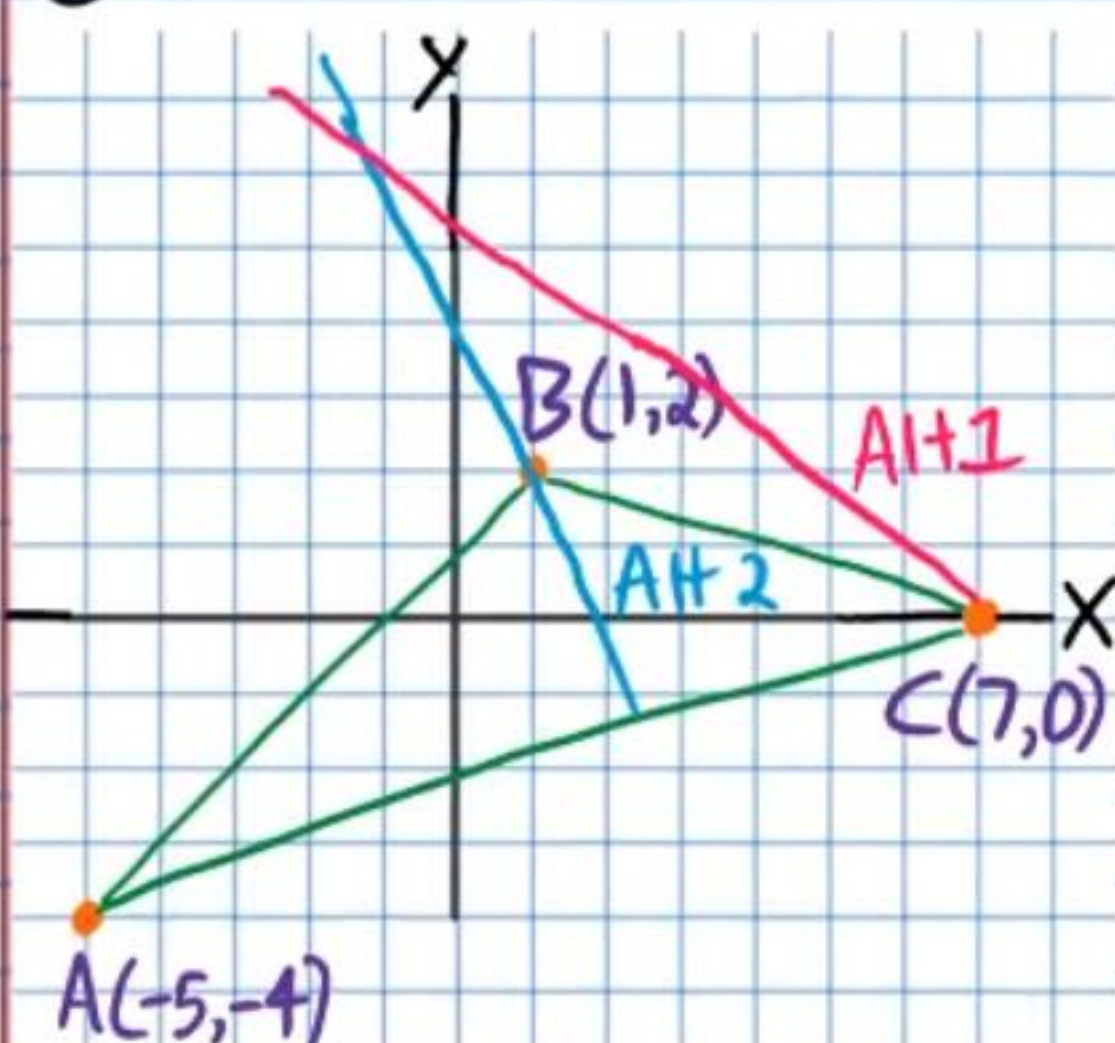
$$\frac{3}{2} = x$$

$$y = -7\left(\frac{3}{2} - 2\right) - 4$$

$$y = -\frac{1}{2}$$

Orthocenter: $\left(\frac{3}{2}, -\frac{1}{2}\right)$

⑦



Line Containing Altitude 1

I) $m_{AB} = \frac{2 - (-4)}{1 - (-5)} = \frac{6}{6} = 1$

II) $m_{\perp AB} = -1$

III) $y - 0 = -(x - 7)$

Line Containing Altitude 2

I) $m_{AC} = \frac{0 - (-4)}{7 - (-5)} = \frac{4}{12} = \frac{1}{3}$

II) $m_{\perp AC} = -3$

III) $y - 2 = -3(x - 1)$

Intersection Point

$$\begin{cases} y - 0 = -(x - 7) \\ y - 2 = -3(x - 1) \end{cases}$$

$$\begin{cases} y = -(x - 7) \\ y - 2 = -3(x - 1) \end{cases}$$

$$-(x - 7) - 2 = -3(x - 1)$$

$$-x + 7 - 2 = -3x + 3$$

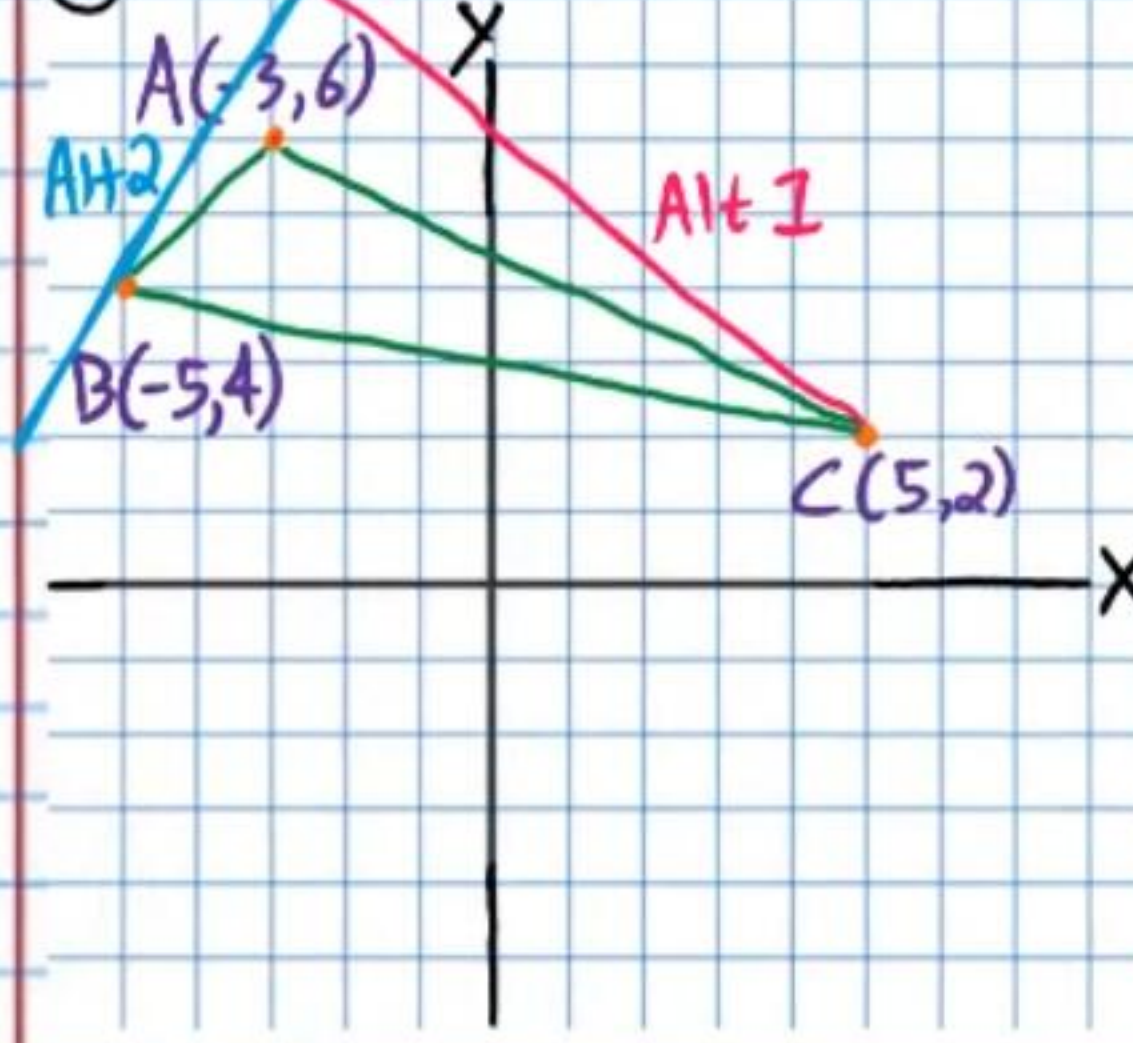
$$2x = -2$$

$$x = -1$$

$$y = -(-1 - 7) = -(-8) = 8$$

Orthocenter: $(-1, 8)$

⑧



Line Containing Altitude 1

I) $m_{AB} = \frac{4 - 6}{-5 - (-3)} = \frac{-2}{-2} = 1$

II) $m_{\perp AB} = -1$

III) $y - 2 = -(x - 5)$

Line Containing Altitude 2

I) $m_{AC} = \frac{2 - 6}{5 - (-3)} = \frac{-4}{8} = -\frac{1}{2}$

II) $m_{\perp AC} = 2$

III) $y - 4 = 2(x + 5)$

Intersection Point

$$\begin{cases} y - 2 = -(x - 5) \\ y - 4 = 2(x + 5) \end{cases}$$

$$\begin{cases} y = -(x - 5) + 2 \\ y - 4 = 2(x + 5) \end{cases}$$

$$-(x - 5) + 2 - 4 = 2(x + 5)$$

$$-x + 5 + 2 - 4 = 2x + 10$$

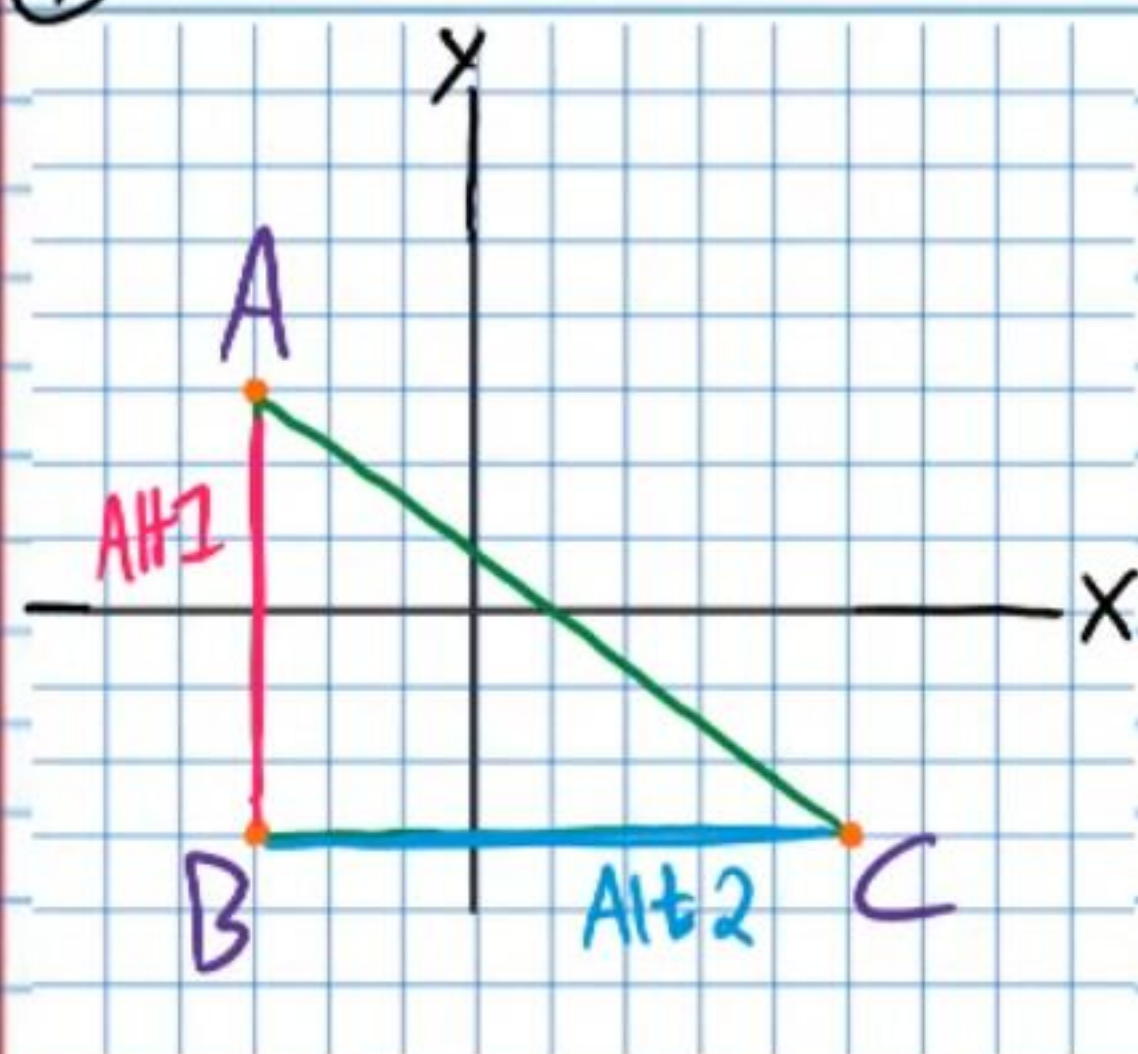
$$-\frac{7}{3} = x$$

$$y = -(-\frac{7}{3} - 5) + 2$$

$$y = \frac{28}{3}$$

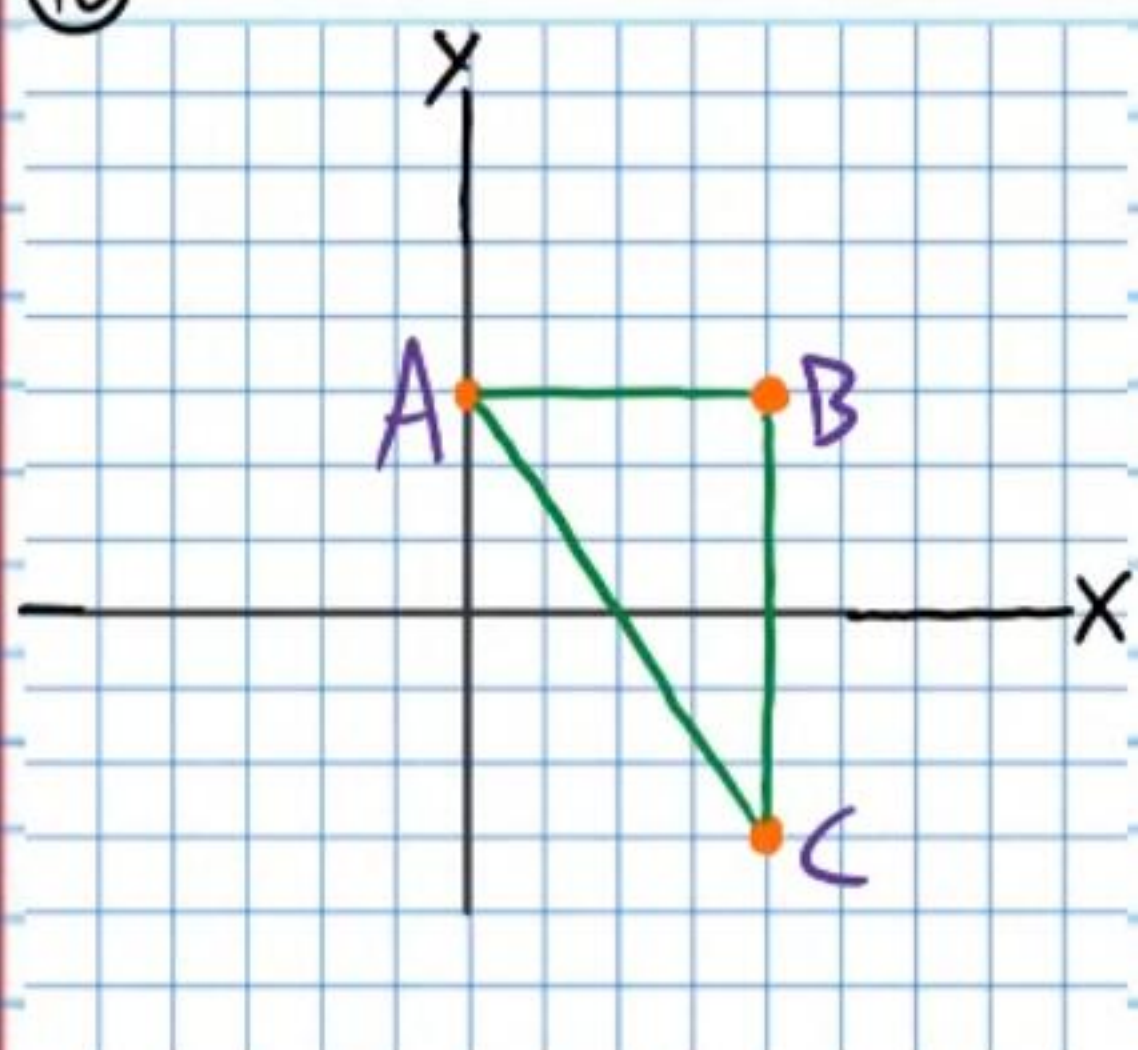
Orthocenter: $(-\frac{7}{3}, \frac{28}{3})$

9



Orthocenter: $(-3, -3)$

10



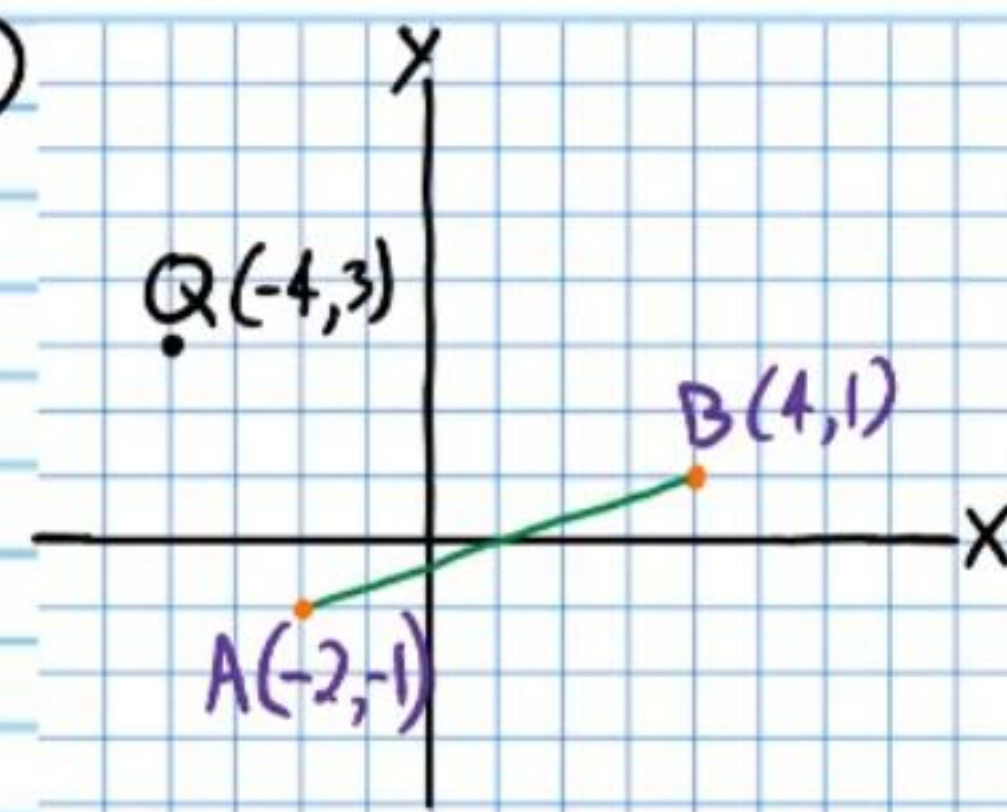
Orthocenter: $(4, 3)$

The Midpoint

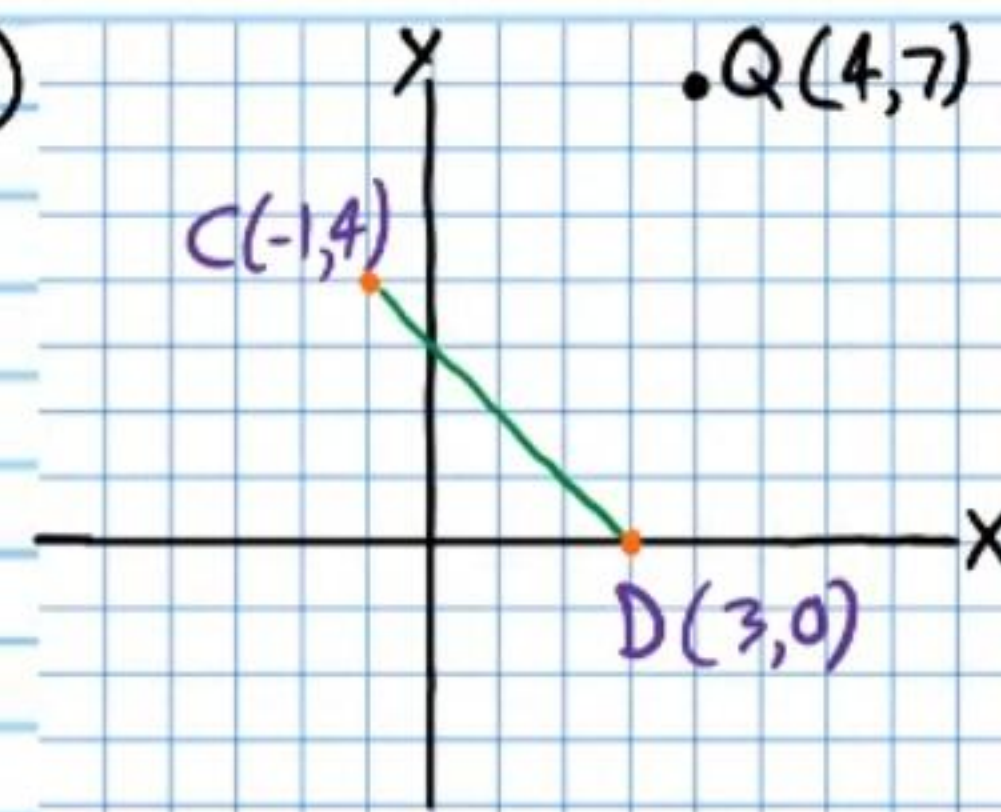
The midpoint between points (x_1, y_1) and (x_2, y_2) is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

Find an equation of the line that runs through point Q and the midpoint of the segment shown.

11



12



I) $M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

$M_{AB}\left(\frac{-2 + 4}{2}, \frac{1 + (-1)}{2}\right)$

$M_{AB}(1, 0)$

II) $M_{MQ} = \frac{3 - 0}{-4 - 1} = -\frac{3}{5}$

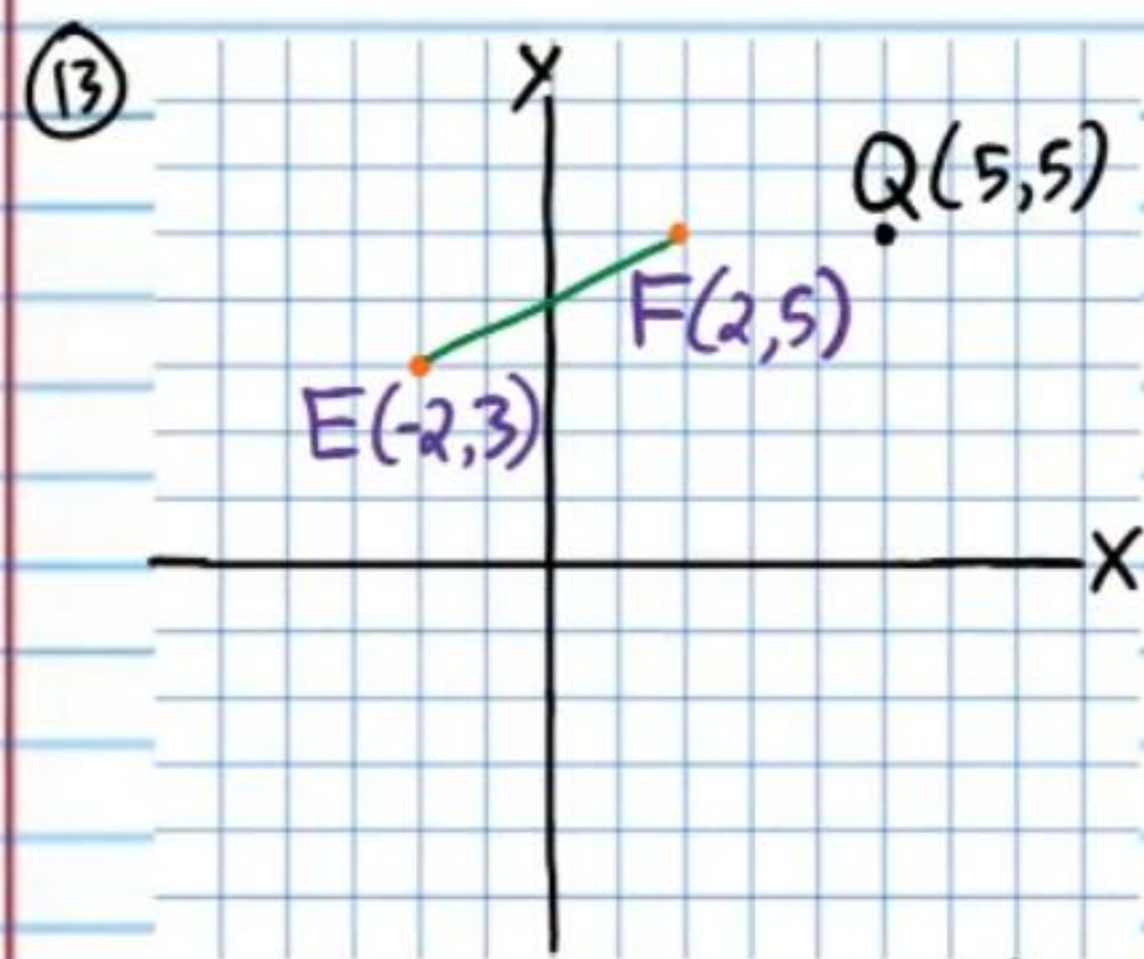
III) $y - 3 = -\frac{3}{5}(x + 4)$

I) $M_{CD}\left(\frac{-1 + 3}{2}, \frac{4 + 0}{2}\right)$

$M_{CD}(1, 2)$

II) $M_{MQ} = \frac{7 - 2}{4 - 1} = \frac{5}{3}$

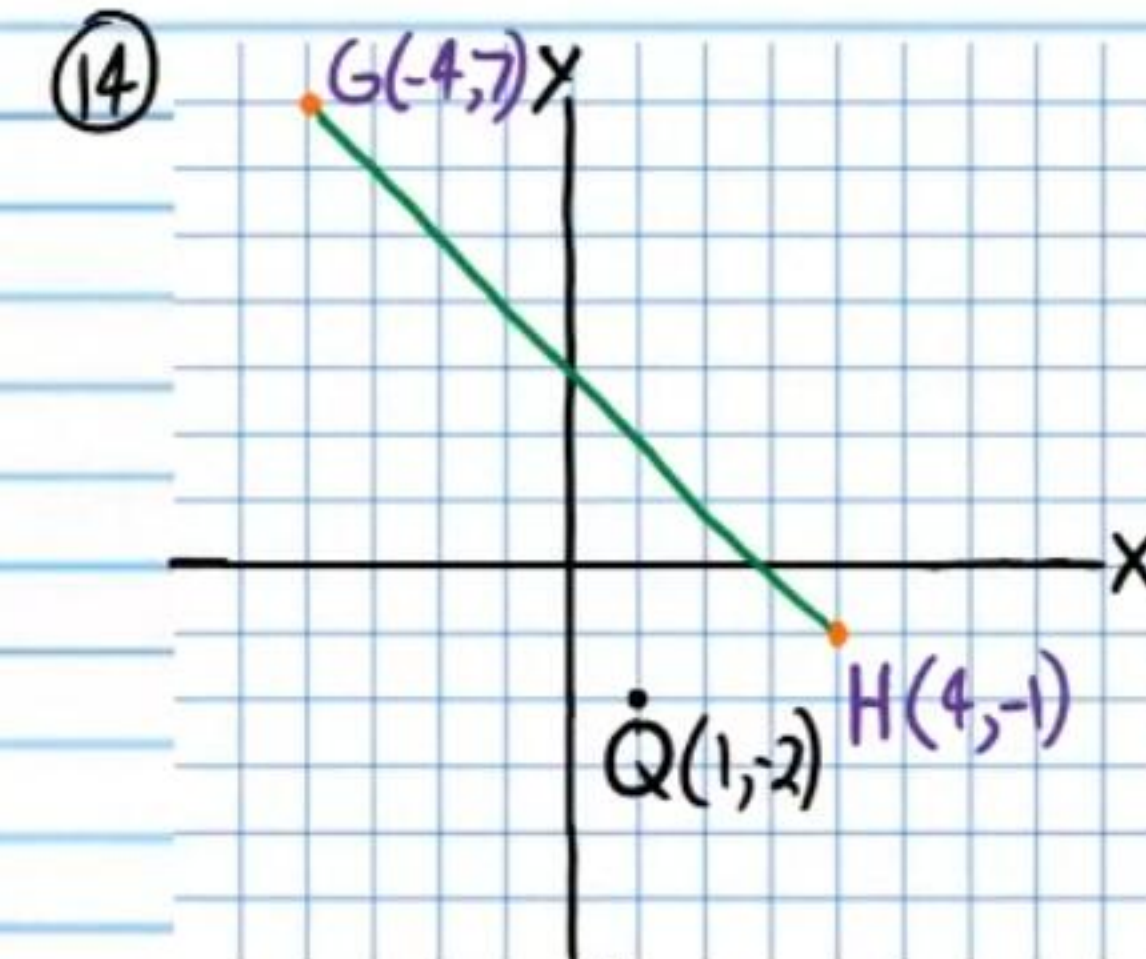
III) $y - 7 = \frac{5}{3}(x - 4)$



I) $M_{EF} \left(\frac{-2+2}{2}, \frac{3+5}{2} \right)$
 $M_{EF} (0, 4)$

II) $M_{MQ} = \frac{5-4}{5-0} = \frac{1}{5}$

III) $y-5 = \frac{1}{5}(x-5)$

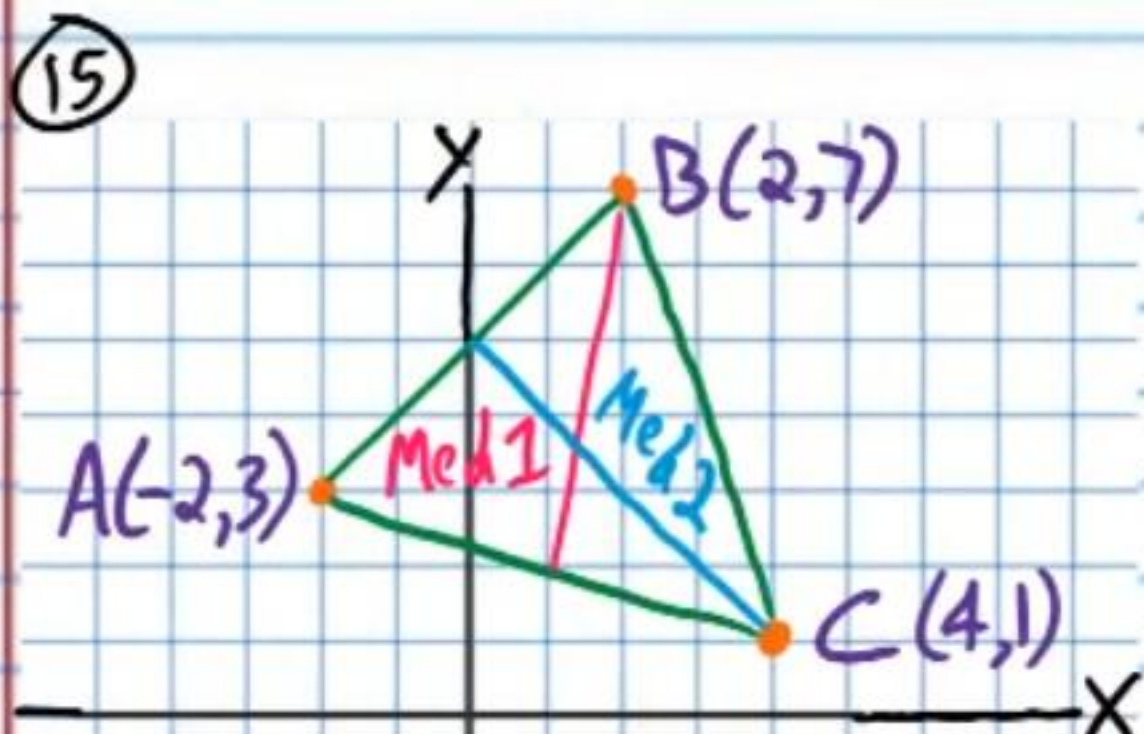


I) $M_{GH} \left(\frac{-4+4}{2}, \frac{7+(-1)}{2} \right)$
 $M_{GH} (0, 3)$

II) $M_{MQ} = \frac{-2-3}{1-0} = \frac{-5}{1} = -5$

III) $y+2 = -5(x-1)$

Given the graphs of the triangles below, find the coordinates of the centroids.



Line Containing Median 1

I) $M_{AC} \left(\frac{-2+4}{2}, \frac{3+1}{2} \right) = M_{AC} (1, 2)$

II) $M_{MB} = \frac{7-2}{2-1} = \frac{5}{1} = 5$

III) $y-7 = 5(x-2)$

Line Containing Median 2

I) $M_{AB} \left(\frac{-2+2}{2}, \frac{3+7}{2} \right) = M_{AB} (0, 5)$

II) $M_{MC} = \frac{1-5}{4-0} = \frac{-4}{4} = -1$

III) $y-1 = -(x-4)$

Intersection Point

$\begin{cases} y-7 = 5(x-2) \\ y-1 = -(x-4) \end{cases}$

$\begin{cases} y = 5(x-2)+7 \\ y-1 = -(x-4) \end{cases}$

$5(x-2)+7-1 = -(x-4)$

$5x-10+7-1 = -x+4$

$5x-4 = -x+4$

$x = \frac{4}{3}$

$y = 5\left(\frac{4}{3}-2\right)+7$

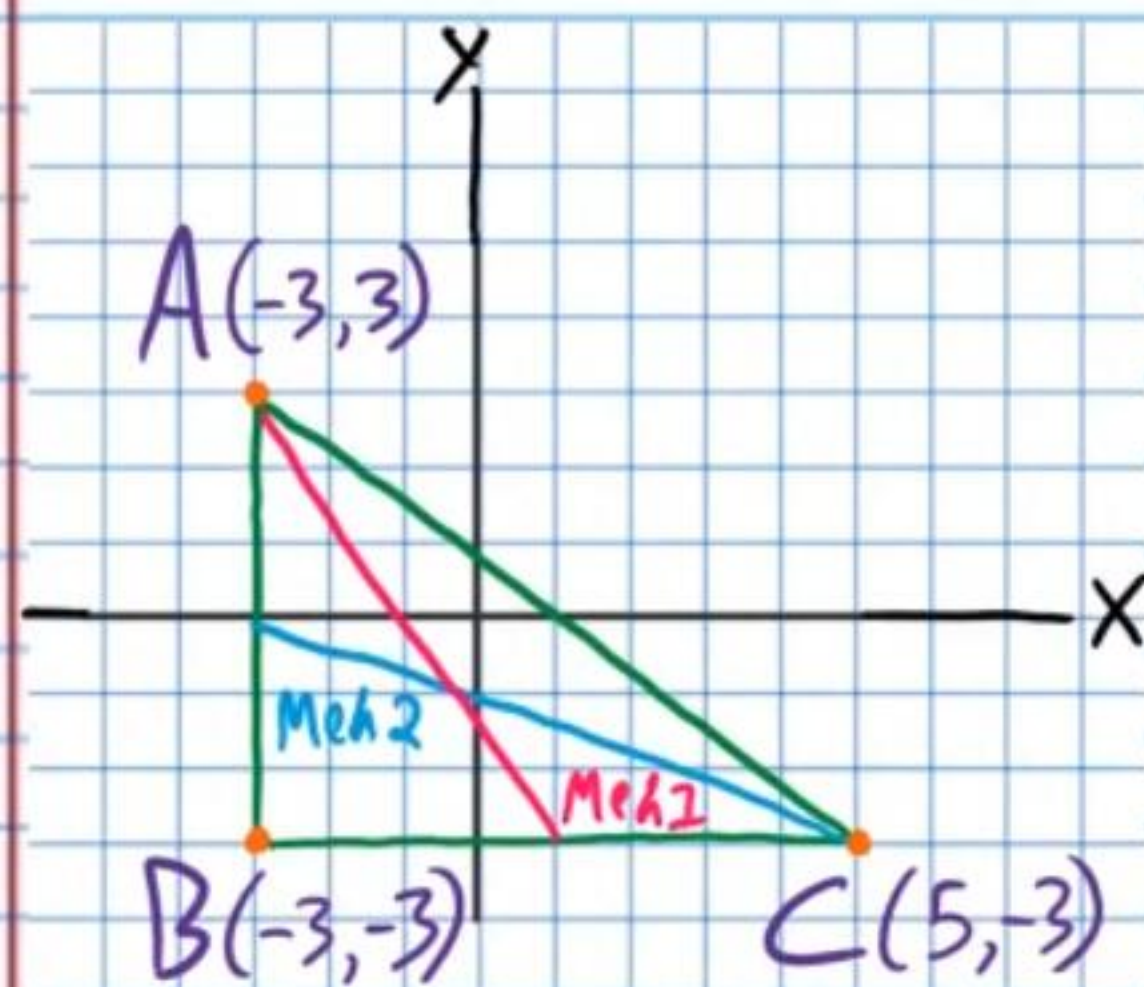
$y = 5\left(-\frac{2}{3}\right)+7$

$y = -\frac{10}{3}+7$

$y = \frac{11}{3}$

Centroid: $\left(\frac{4}{3}, \frac{11}{3}\right)$

(16)

Line Containing Median 1

I) $M_{BC}(1, -3)$

II) $M_{MA} = \frac{3 - (-3)}{-3 - 1} = \frac{6}{-4} = -\frac{3}{2}$

III) $y - 3 = -\frac{3}{2}(x + 3)$

Line Containing Median 2

I) $M_{AB}(-3, 0)$

II) $M_{MC} = \frac{-3 - 0}{5 - (-3)} = -\frac{3}{8}$

III) $y + 3 = -\frac{3}{8}(x - 5)$

Intersection Point

$$\begin{cases} y - 3 = -\frac{3}{2}(x + 3) \\ y + 3 = -\frac{3}{8}(x - 5) \end{cases}$$

$$\begin{cases} y = -\frac{3}{2}(x + 3) + 3 \\ y + 3 = -\frac{3}{8}(x - 5) \end{cases}$$

$$\begin{cases} y = -\frac{3}{2}(x + 3) + 3 \\ y + 3 = -\frac{3}{8}(x - 5) \end{cases}$$

$$\begin{cases} y = -\frac{3}{2}(x + 3) + 3 \\ y + 3 = -\frac{3}{8}(x - 5) \end{cases}$$

$$-\frac{3}{2}(x + 3) + 3 + 3 = -\frac{3}{8}(x - 5)$$

$$-\frac{3}{2}x - \frac{9}{2} + 3 + 3 = -\frac{3}{8}x + \frac{15}{8}$$

$$-\frac{3}{2}x + \frac{3}{2} = -\frac{3}{8}x + \frac{15}{8}$$

$$\frac{-3x + 3}{2} = \frac{-3x + 15}{8}$$

$$\frac{4(-3x + 3)}{4(2)} = \frac{-3x + 15}{8}$$

$$\frac{-12x + 12}{8} = \frac{-3x + 15}{8}$$

$$-12x + 12 = -3x + 15$$

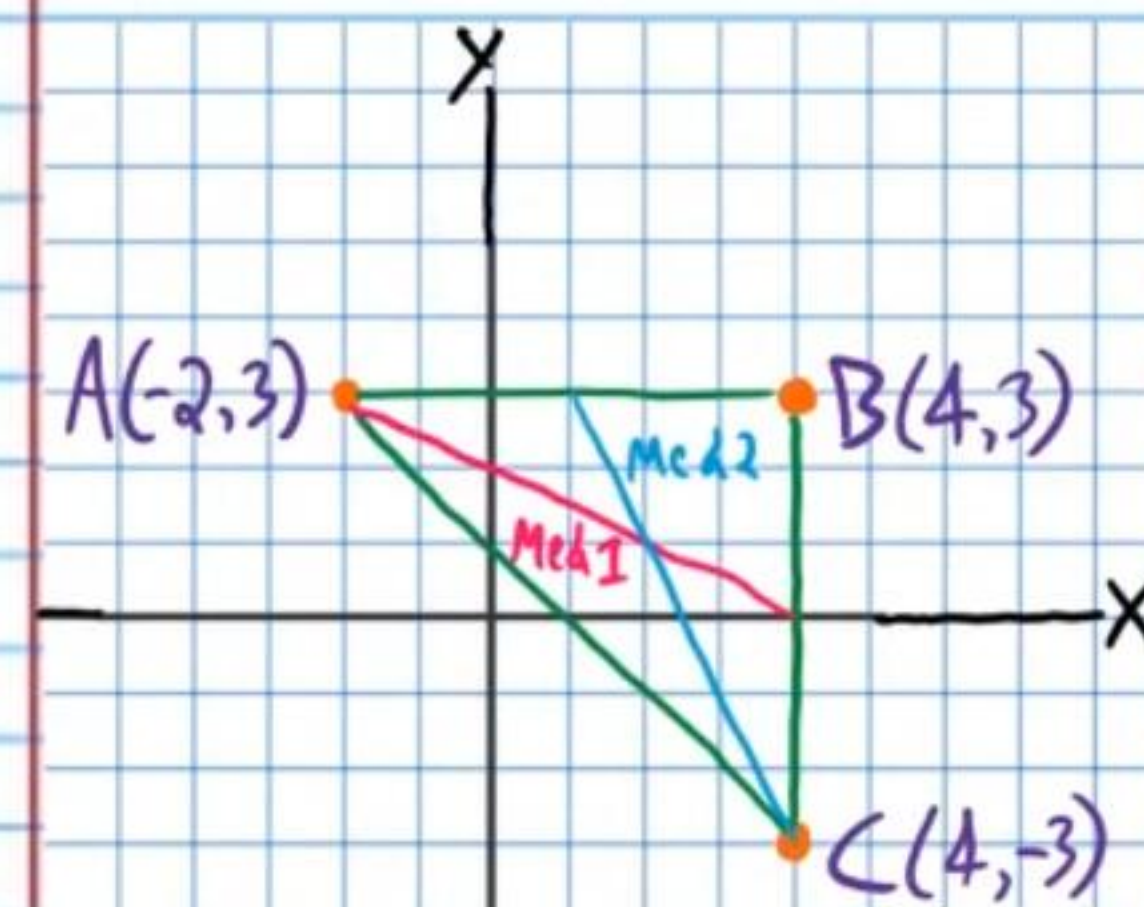
$$-\frac{1}{3} = x$$

$$y = -\frac{3}{2}\left(-\frac{1}{3} + 3\right) + 3$$

$$y = -1$$

$$\text{Centroid: } \left(-\frac{1}{3}, -1\right)$$

(17)

Line Containing Median 1

I) $M_{BC}(4, 0)$

II) $M_{MA} = \frac{3 - 0}{-2 - 4} = \frac{3}{-6} = -\frac{1}{2}$

III) $y - 3 = -\frac{1}{2}(x + 2)$

Line Containing Median 2

I) $M_{AB}(1, 3)$

II) $M_{MC} = \frac{-3 - 3}{4 - 1} = \frac{-6}{3} = -2$

III) $y + 3 = -2(x - 4)$

Intersection Point

$$\begin{cases} y - 3 = -\frac{1}{2}(x + 2) \\ y + 3 = -2(x - 4) \end{cases}$$

$$\begin{cases} y - 3 = -\frac{1}{2}(x + 2) \\ y + 3 = -2(x - 4) \end{cases}$$

$$\begin{cases} y = -\frac{1}{2}(x + 2) + 3 \\ y + 3 = -2(x - 4) \end{cases}$$

$$\begin{cases} y = -\frac{1}{2}(x + 2) + 3 \\ y + 3 = -2(x - 4) \end{cases}$$

$$-\frac{1}{2}(x + 2) + 3 + 3 = -2(x - 4)$$

$$-\frac{1}{2}x - 1 + 3 + 3 = -2x + 8$$

$$-\frac{1}{2}x + 5 = -2x + 8$$

$$-\frac{1}{2}x + \frac{10}{2} = \frac{-4x + 16}{2}$$

$$\frac{-x + 10}{2} = \frac{-4x + 16}{2}$$

$$-x + 10 = -4x + 16$$

$$-x + 10 = -4x + 16$$

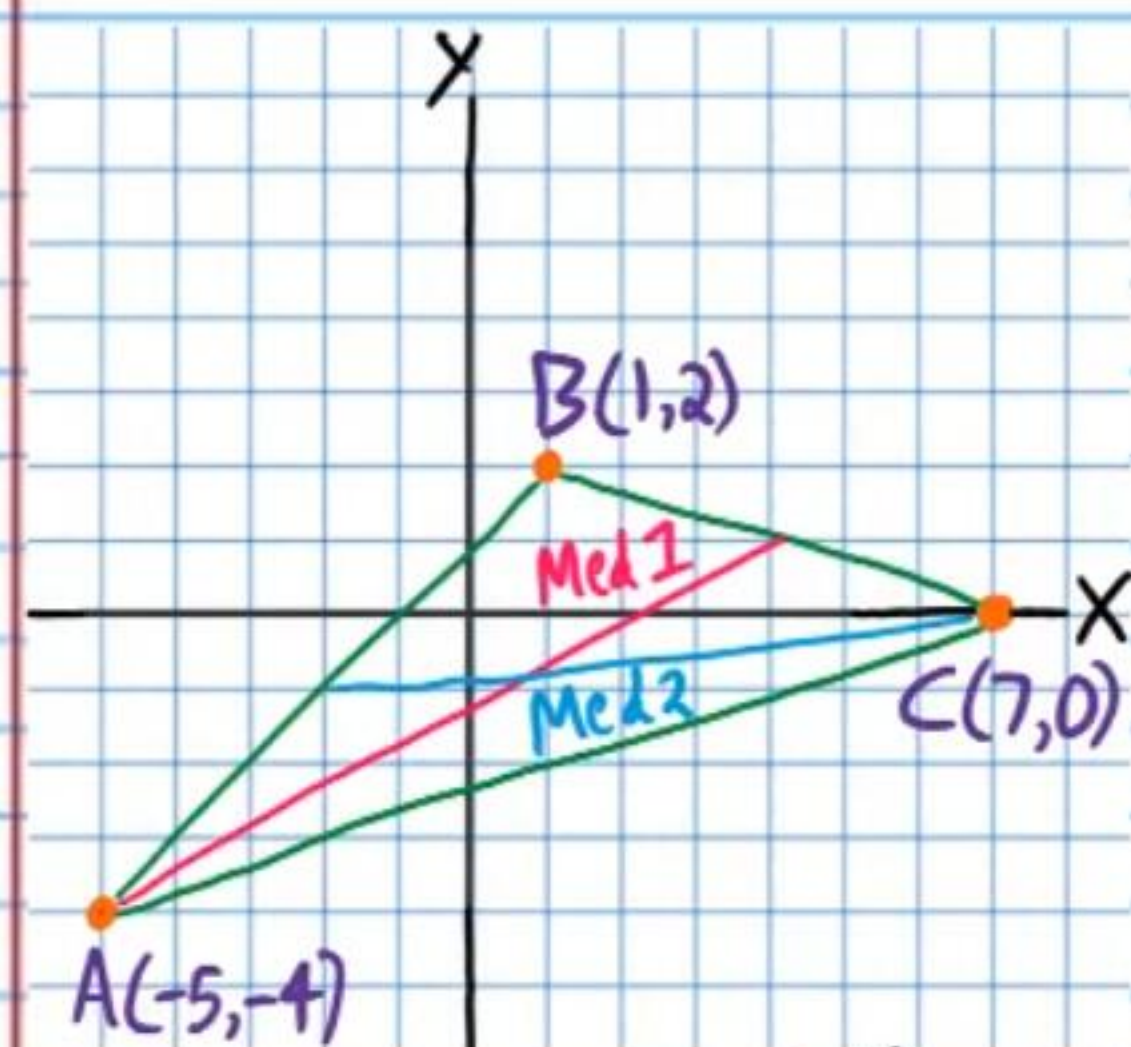
$$x = 2$$

$$y = -\frac{1}{2}(2 + 2) + 3$$

$$y = 1$$

$$\text{Centroid: } (2, 1)$$

18)



Line Containing Median 1

I) $M_{BC}(\frac{1+7}{2}, \frac{2+0}{2}) = M_{BC}(4, 1)$

II) $M_{MA} = \frac{-4-1}{-5-4} = \frac{-5}{-9} = \frac{5}{9}$

III) $y+4 = \frac{5}{9}(x+5)$

Line Containing Median 2

I) $M_{AB}(\frac{1+(-5)}{2}, \frac{2+(-4)}{2})$

$M_{AB}(-2, -1)$

II) $M_{MC} = \frac{0-(-1)}{7-(-2)} = \frac{1}{9}$

III) $y-0 = \frac{1}{9}(x-7)$

Intersection Point

$$\begin{cases} y+4 = \frac{5}{9}(x+5) \\ y-0 = \frac{1}{9}(x-7) \end{cases}$$

$$\begin{cases} y+4 = \frac{5}{9}(x+5) \\ y = \frac{1}{9}(x-7) \end{cases}$$

$$\frac{1}{9}(x-7) + 4 = \frac{5}{9}(x+5)$$

$$\frac{1}{9}x - \frac{7}{9} + \frac{4}{1} = \frac{5}{9}x + \frac{25}{9}$$

$$\frac{x}{9} + \frac{29}{9} = \frac{5x}{9} + \frac{25}{9}$$

$$\frac{x+29}{9} = \frac{5x+25}{9}$$

$$x+29 = 5x+25$$

$$1 = x$$

$$y = \frac{1}{9}(1-7)$$

$$y = \frac{1}{9}(-6)$$

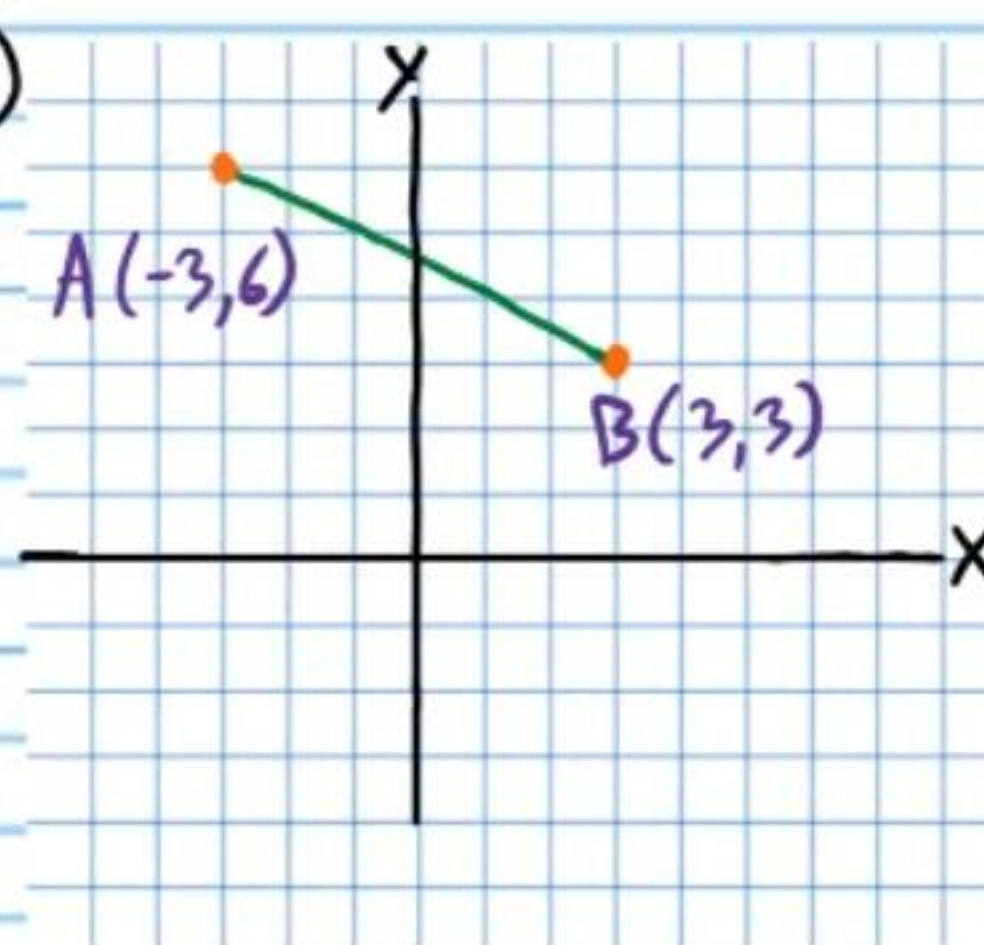
$$y = -\frac{6}{9}$$

$$y = -\frac{2}{3}$$

Centroid: $(1, -\frac{2}{3})$

For the problems below, find an equation of the line which is a perpendicular bisector of the segment graphed.

19)



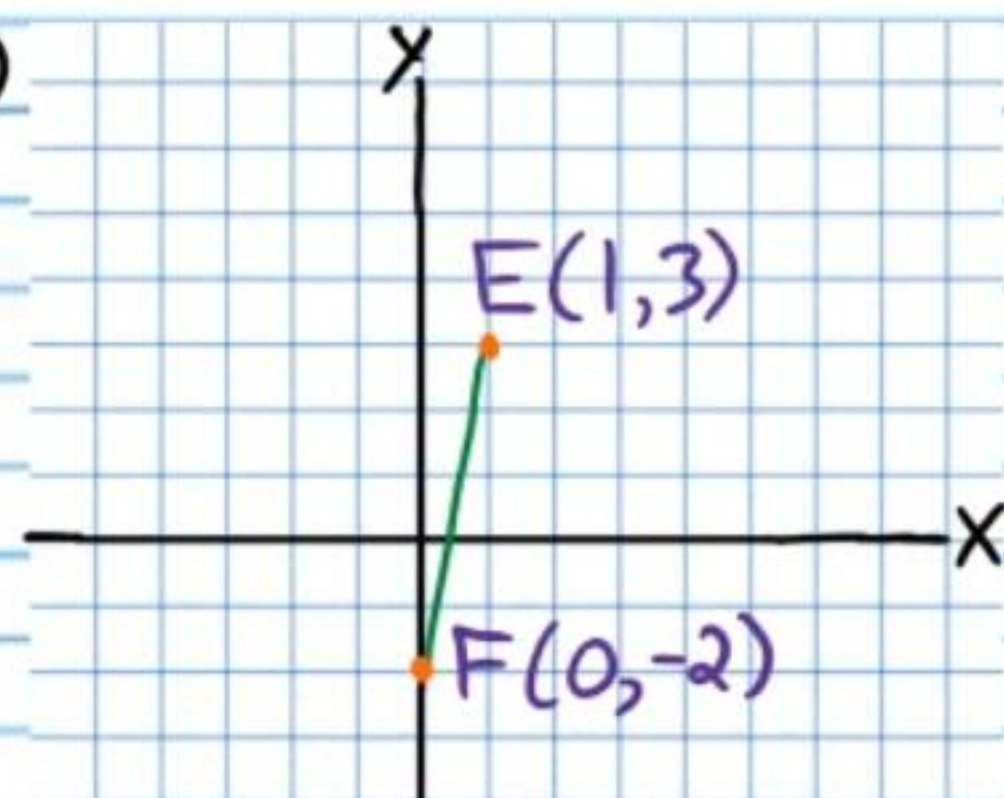
I) $M_{AB}(\frac{-3+3}{2}, \frac{6+3}{2}) = M_{AB}(0, \frac{9}{2})$

II) $M_{AB} = \frac{3-6}{3-(-3)} = \frac{-3}{6} = -\frac{1}{2}$

III) $M_{\perp AB} = 2$

IV) $y - \frac{9}{2} = 2(x - 0)$

(20)



$$\text{I) } M_{EF} \left(\frac{1+0}{2}, \frac{3+(-2)}{2} \right)$$

$$M_{EF} \left(\frac{1}{2}, \frac{1}{2} \right)$$

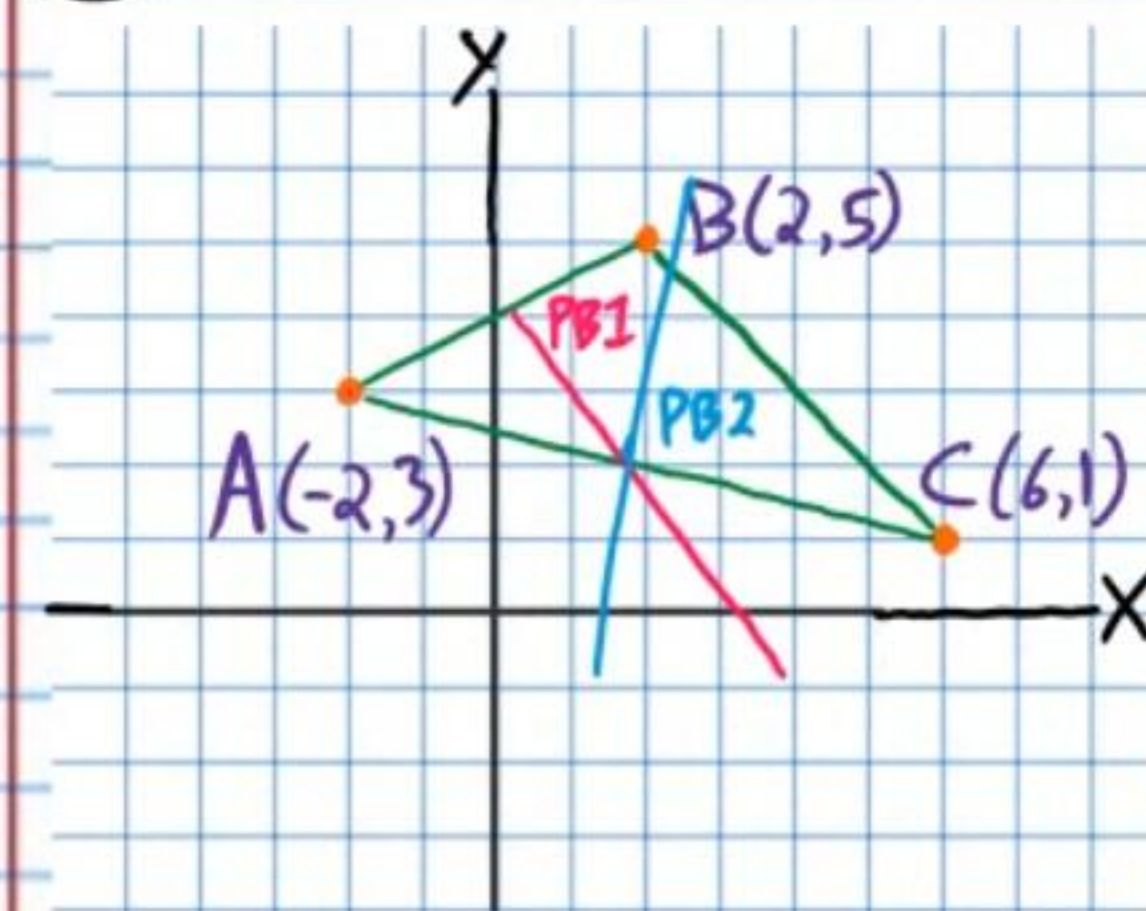
$$\text{II) } M_{EF} = \frac{-2-3}{0-1} = \frac{-5}{-1} = 5$$

$$\text{III) } M_{\perp EF} = -\frac{1}{5}$$

$$\text{IV) } \boxed{y - \frac{1}{2} = -\frac{1}{5} \left(x - \frac{1}{2} \right)}$$

Given the graphs of the triangles below, find the coordinates of the circumcenters.

(21)



Line Containing PB 1

$$\text{I) } M_{AB} \left(\frac{2+(-2)}{2}, \frac{3+5}{2} \right) = M_{AB}(0, 4)$$

$$\text{II) } M_{AB} = \frac{5-3}{2-(-2)} = \frac{2}{4} = \frac{1}{2}$$

$$\text{III) } M_{\perp AB} = -2$$

$$\text{IV) } \boxed{y - 4 = -2(x - 0)}$$

Line Containing PB 2

$$\text{I) } M_{AC} \left(\frac{-2+6}{2}, \frac{3+1}{2} \right) = M_{AC}(2, 2)$$

$$\text{II) } M_{AC} = \frac{1-3}{6-(-2)} = \frac{-2}{8} = -\frac{1}{4}$$

$$\text{III) } M_{\perp AC} = 4$$

$$\text{IV) } \boxed{y - 2 = 4(x - 2)}$$

Intersection Point

$$\begin{cases} y - 4 = -2(x - 0) \\ y - 2 = 4(x - 2) \end{cases}$$

$$\begin{cases} y = -2(x - 0) + 4 \\ y - 2 = 4(x - 2) \end{cases}$$

$$\begin{cases} y = -2(x - 0) + 4 \\ y - 2 = 4(x - 2) \end{cases}$$

$$-2(x - 0) + 4 - 2 = 4(x - 2)$$

$$-2x + 4 - 2 = 4x - 8$$

$$-2x + 2 = 4x - 8$$

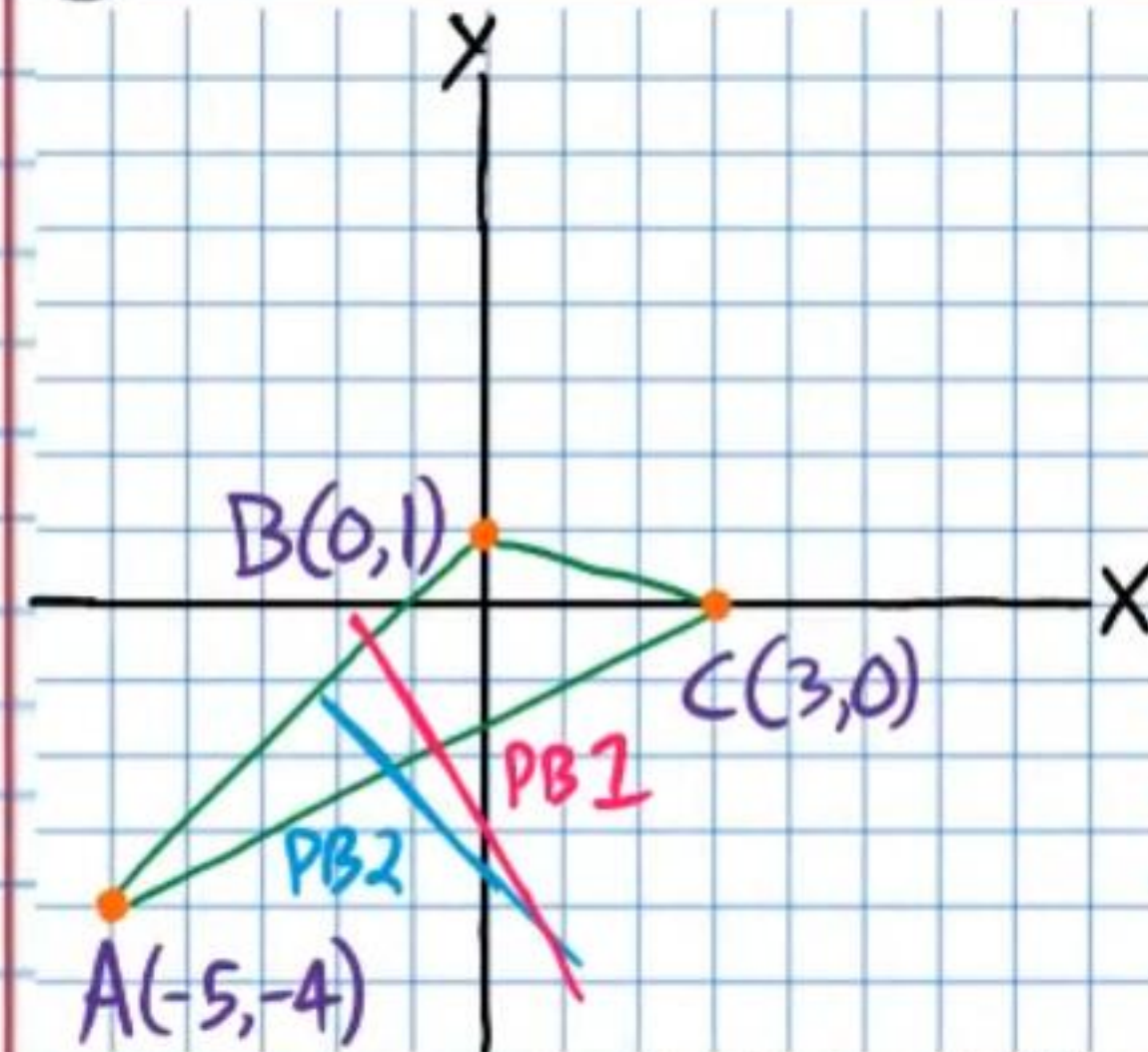
$$\frac{5}{3} = x$$

$$y = -2 \left(\frac{5}{3} - 0 \right) + 4$$

$$y = \frac{2}{3}$$

$$\boxed{\text{Circumcenter: } \left(\frac{5}{3}, \frac{2}{3} \right)}$$

22



Line Containing PB 1

$$I) M_{AC} = \left(\frac{-5+3}{2}, \frac{-4+0}{2} \right) = M_{AC}(-1, -2)$$

$$II) M_{AC} = \frac{0 - (-4)}{3 - (-5)} = \frac{4}{8} = \frac{1}{2}$$

$$III) M_{\perp AC} = -2$$

$$IV) y + 2 = -2(x + 1)$$

Line Containing PB 2

$$I) M_{AB} = \left(\frac{0+(-5)}{2}, \frac{1+(-4)}{2} \right) = M_{AB}\left(-\frac{5}{2}, -\frac{3}{2}\right)$$

$$II) M_{AB} = \frac{1 - (-4)}{0 - (-5)} = \frac{5}{5} = 1$$

$$III) M_{\perp AB} = -1$$

$$IV) y + \frac{3}{2} = -(x + \frac{5}{2})$$

Intersection Point

$$\begin{cases} y + 2 = -2(x + 1) \\ y + \frac{3}{2} = -(x + \frac{5}{2}) \end{cases}$$

$$\begin{cases} y = -2(x + 1) - 2 \\ y + \frac{3}{2} = -(x + \frac{5}{2}) \end{cases}$$

$$\rightarrow -2(x + 1) - 2 + \frac{3}{2} = -(x + \frac{5}{2})$$

$$-2x - 2 - 2 + \frac{3}{2} = -x - \frac{5}{2}$$

$$-2x - \frac{5}{2} = -x - \frac{5}{2}$$

$$-2x = -x$$

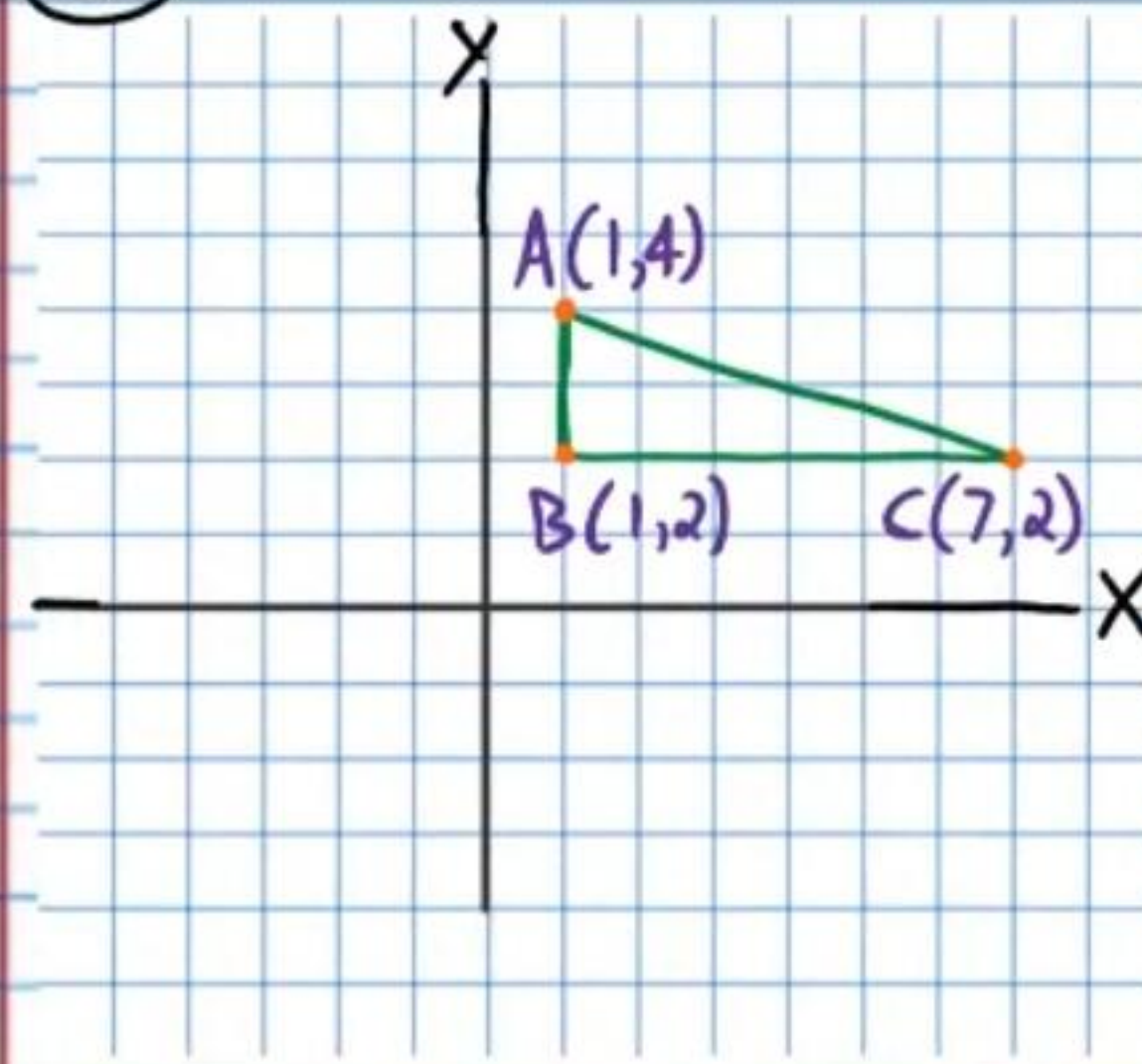
$$0 = x$$

$$y = -2(0 + 1) - 2$$

$$y = -4$$

$$\text{Circumcenter: } (0, -4)$$

23

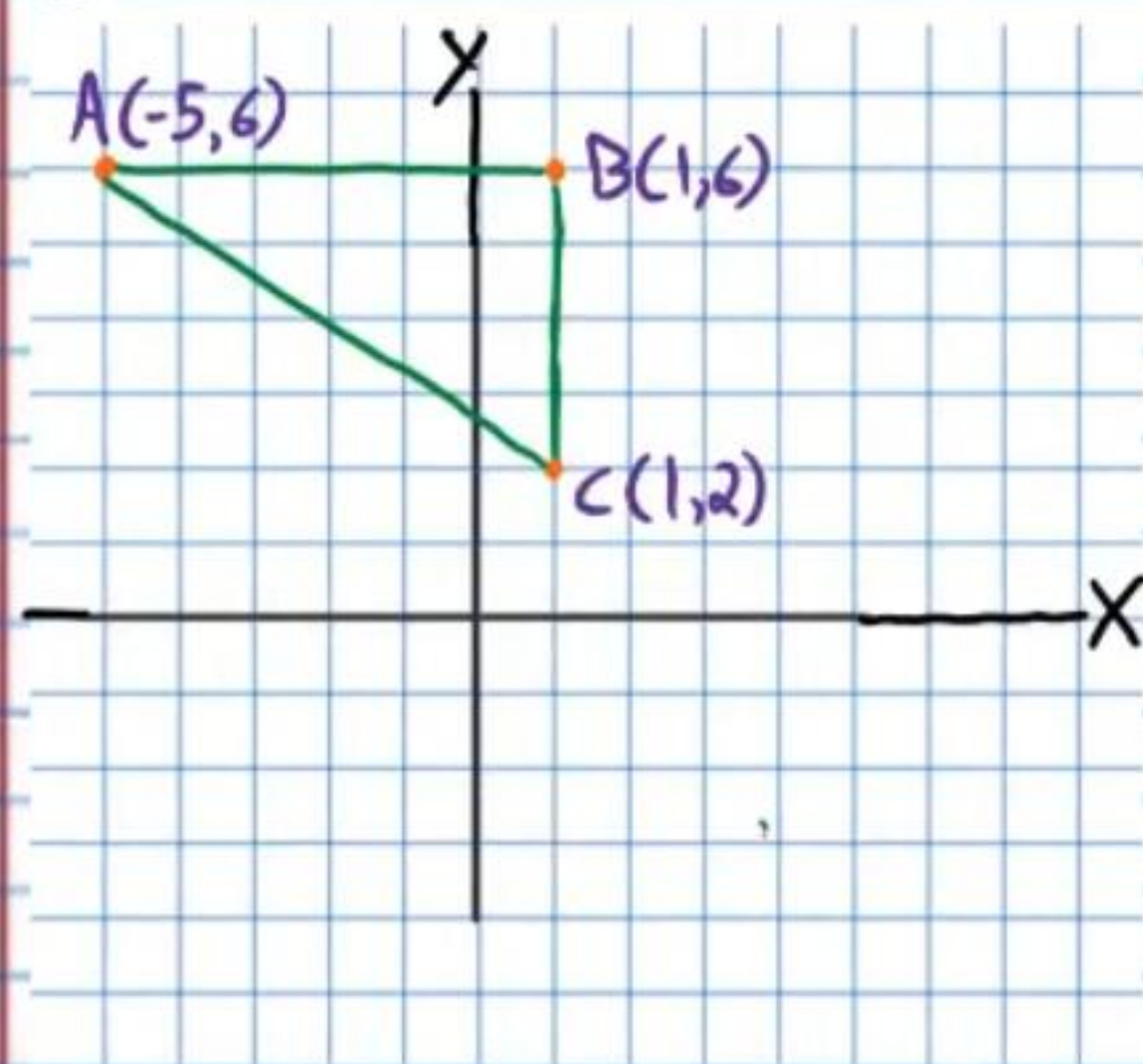


$$x = 4$$

$$y = 3$$

$$\text{Circumcenter: } (4, 3)$$

24



$$x = -2$$

$$y = 4$$

$$\text{Circumcenter: } (-2, 4)$$